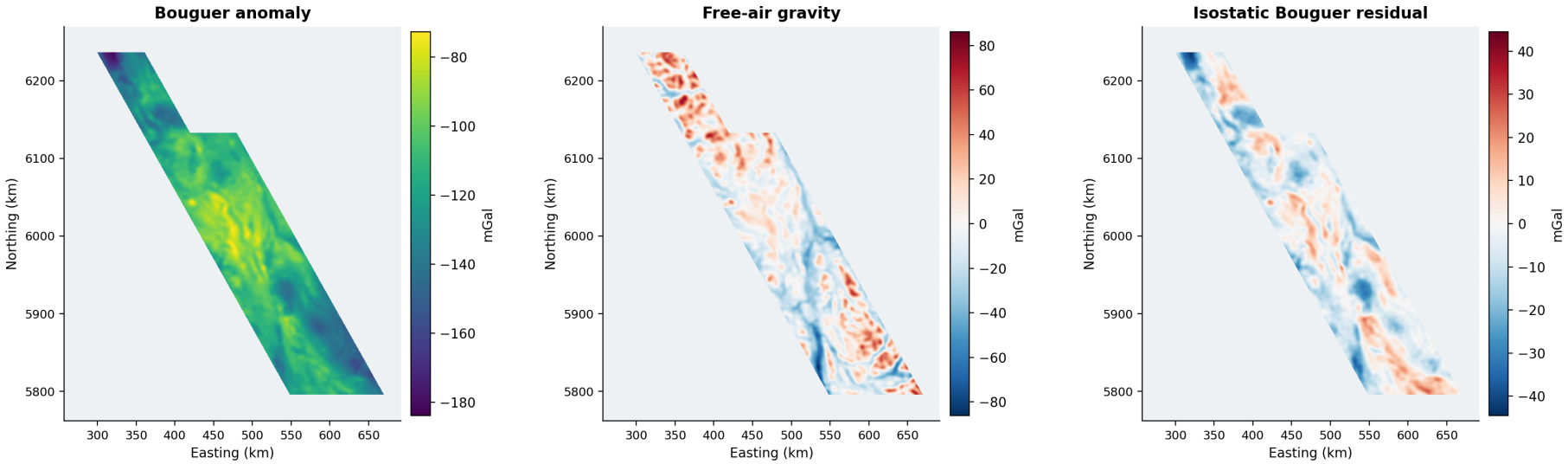
QUEST airborne gravity

2D anomaly mapping, metadata auditing and held-line interpolation validation | Original QUEST survey, British Columbia

OBSERVATIONS	LINE SECTIONS	LINEAR CV RMSE	SUPPORTED TEST BLOCKS
1,049,547	340	1.912 mGal	98.8%

QUEST | Regional airborne gravity

Published fields converted in Python | 250 m grid nodes; supplier 3.5 km half-wavelength filtering | Values in mGal



Source: Geoscience BC Report 2008-08 / Sander Geophysics | NAD83 / UTM 10N (EPSG:26910)

Outcome. The supplier fields were converted into GIS-ready rasters, and a separate line-data interpolation was tested with whole planned lines withheld. The 250 m source-grid spacing is not a claim of 250 m geophysical resolution.

Audit before interpretation

The release contains conflicting metadata. Each decision is recorded and checked; the original source files remain unchanged.

Issue	Decision and measured evidence
CRS conflict	README/XYZ specify NAD83 / UTM 10N; report table says zone 32N. Use EPSG:26910. A 10,496-point coordinate check gives a median discrepancy of 1.303 m, supporting the zone selection. This is not positional accuracy.
File format	Released grids are binary Geosoft files despite an ASCII GXF label in the report. A guarded reader accepts only the observed encoding. Actual grid-versus-line sampling checks values and orientation.
Gravity channel labels	Some final channel descriptions incorrectly suggest isostatic correction on every field. Grid correspondence and the isostatic-channel identity support treating the three products separately. Do not apply corrections twice.
Correction history	The line table says simple Bouguer, while the narrative includes terrain and curvature. GRVBGL100 is labelled as the supplier Bouguer channel without claiming a newly computed correction decomposition.
Time and missing values	43,663 records have SECOND >= 86,400 and require rollover interpretation for time-based work. No missing numeric values or duplicate flight/line/time records were found. Spatial analysis retains these observations.

Source consistency checks

Supplier grid / line channel	RMSE of related products (mGal)
GRVBG3.5KM.grd / GRVBGL3.5	0.00446
GRVFA3.5KM.grd / GRVFRAL3.5	0.01108
ISOBG3.5KM.grd / ISOBGL3.5	0.00447

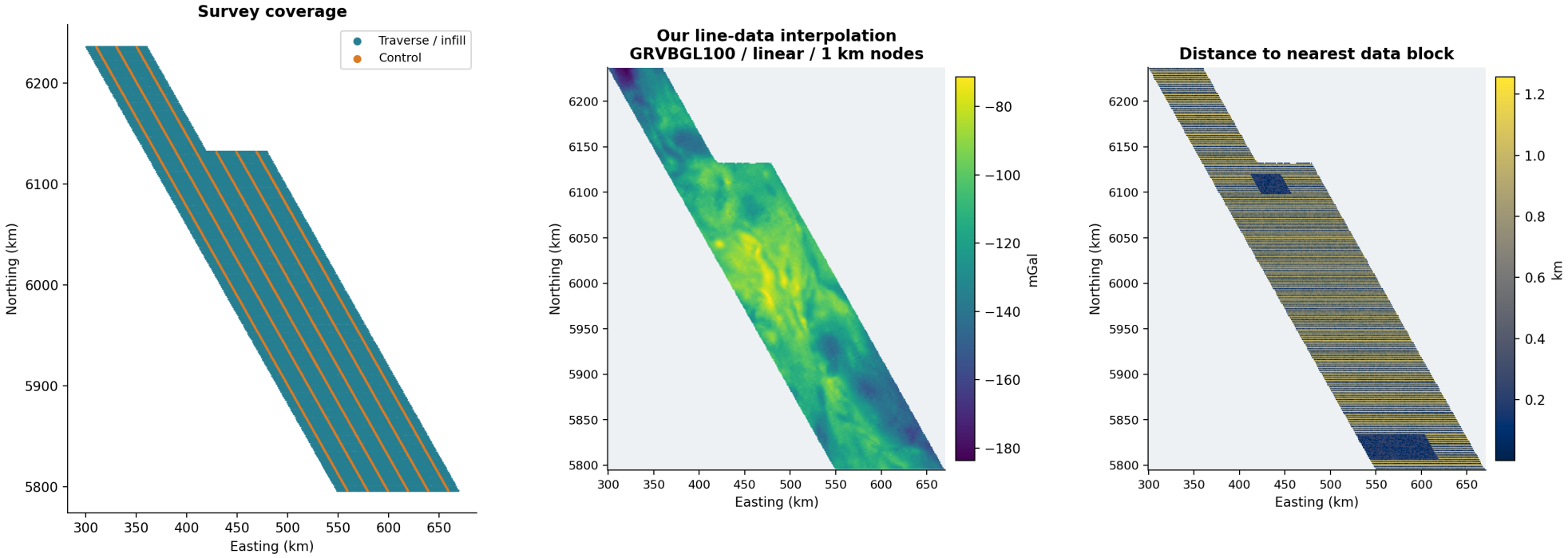
These tiny discrepancies verify correspondence of related products. They are not independent validation against ground truth. Full evidence and unresolved details are in docs/METADATA_ISSUES.md.

Line observations to supported rasters

A second workflow uses GRVBGL100, preserving a clear distinction from the supplier's final spatially filtered grids.

QUEST | From line observations to GIS rasters

500 m per-line block means | Outside-hull, long-triangle and distance masking | No extrapolation



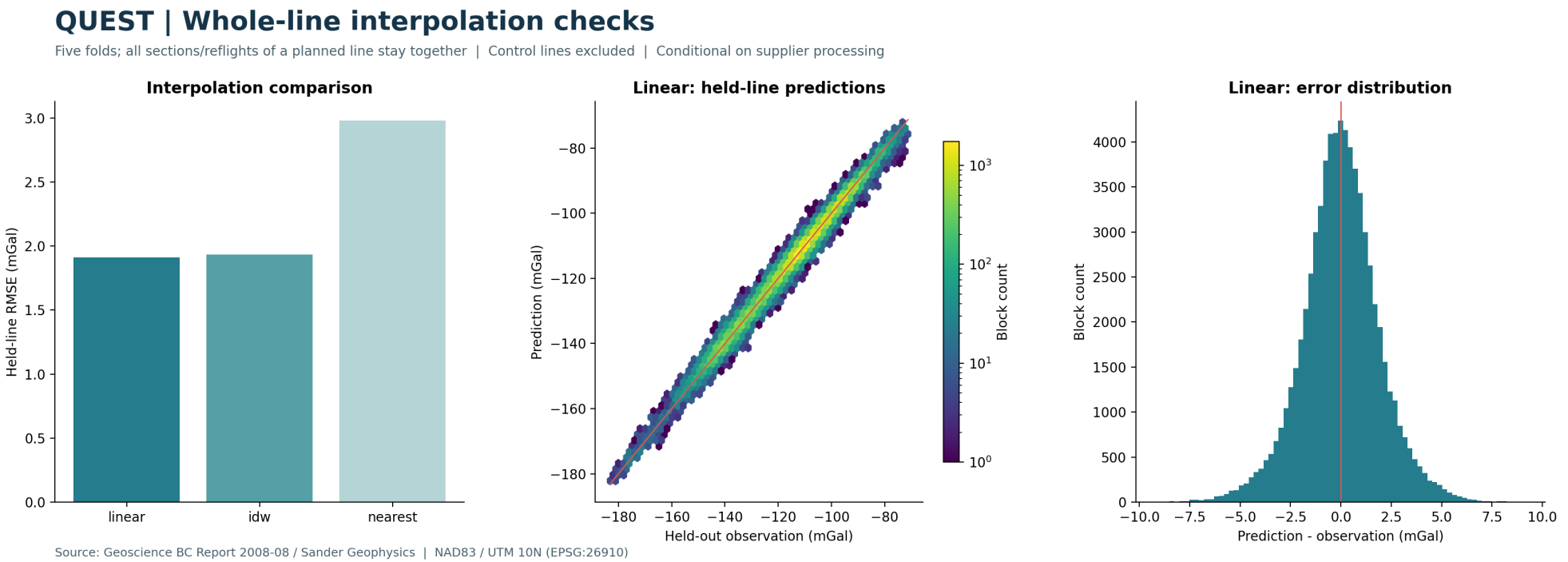
Source: Geoscience BC Report 2008-08 / Sander Geophysics | NAD83 / UTM 10N (EPSG:26910)

Method. Average samples in 500 m XY cells separately for each exact line section. Use traverse/infill lines for linear triangulation; exclude control lines. Export a 1 km node raster. Mask outside the convex hull, beyond 3 km from data or inside triangles with edges longer than 7 km.

Interpretation. Distance to data is a support diagnostic, not a formal uncertainty estimate. These masks limit gap bridging. Linear interpolation can leave triangular features; a visually detailed surface is not evidence of finer geophysical resolving power.

Whole-line validation

Five interleaved folds; all sections and reflights of a planned line remain together. Control lines are excluded from training and testing.



Interpolator	RMSE (mGal)	MAE (mGal)	Bias (mGal)	Coverage
linear	1.912	1.455	-0.011	98.8%
idw	1.934	1.476	-0.007	98.8%
nearest	2.981	2.273	-0.002	98.8%

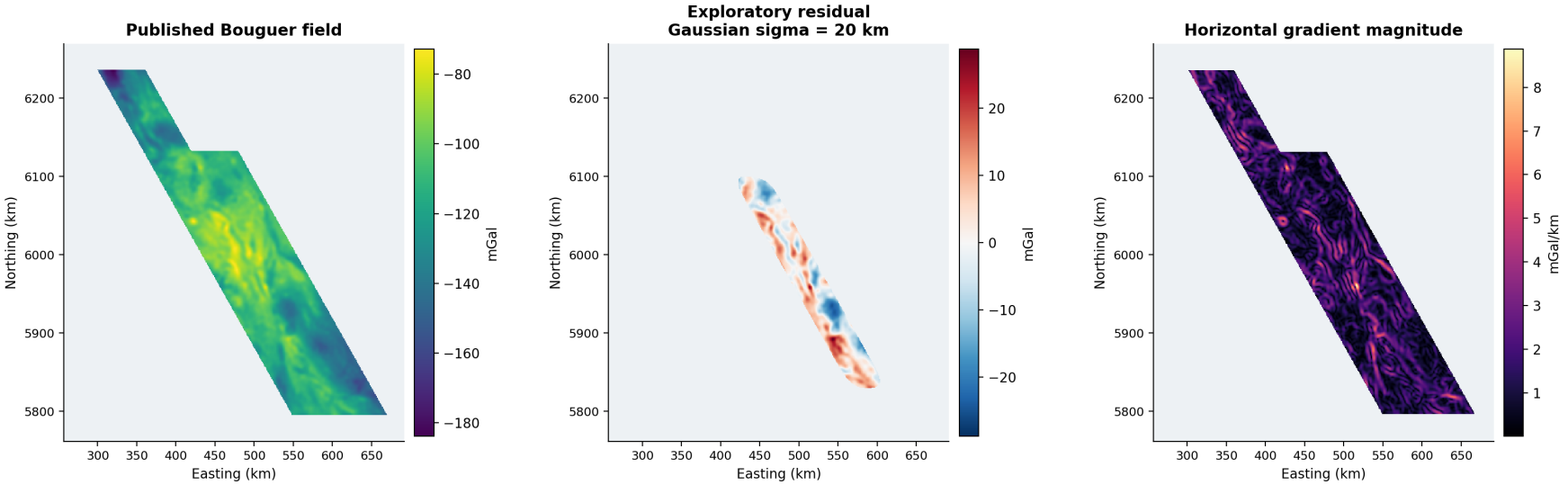
Limit. The errors describe interpolation of already corrected and levelled data. Supplier-wide levelling, nearby infill lines and spatial correlation remain. No independent instrument accuracy or operational navigation performance is inferred. Linear was fixed in advance; the small difference versus IDW is not a statistical-significance claim.

Spatial analysis and the next experiment

Exploratory regional/residual separation and horizontal gradients connect this study to questions about map structure.

QUEST | Wavelengths and spatial structure

Residual requires 95% kernel support | Gradient is a map-texture indicator, not a positioning-accuracy estimate



Source: Geoscience BC Report 2008-08 / Sander Geophysics | NAD83 / UTM 10N (EPSG:26910)

Scale and edges. The residual subtracts a Gaussian regional component with sigma = 20 km and requires 95% kernel support. On a common 11,969-cell area, residual standard deviation changes from 6.20 to 7.99 to 9.22 mGal for sigma = 10, 15 and 20 km. No cells support sigma = 30 km at the chosen threshold.

Navigation relevance. Gradients describe spatial variation, not unique position observability. Before a navigation simulation, agree on the gravity functional, observation height, terrain/long-wavelength treatment, independent validation data and realistic sensor/map errors.

Reproduce: README.md, config.json, requirements-lock.txt and outputs/metadata.json. Eight numerical/GIS tests passed. AI-assisted initial implementation; candidate review and independent rerun are recommended before presentation. Source: geosciencebc.com/major-projects/quest/.